Controlled E-Commerce Product Image Generation

Quantitative Evaluation Report — CS 5542 Quiz Challenge 1

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60	0.442	+22.8%	0.279
TOTAL IMAGES GENERATED	AVG SSIM (IMPROVED)	SSIM IMPROVEMENT	AVG CLIP (IMPROVED)
10 products × 3 seeds × 2 prompts	vs 0.360 baseline	Cross-seed consistency gain	Subject recognition preserved

1. Methodology

1.1 Prompt Strategies

BASELINE (Naive)

Uses **title only.**

Example: "*Coffee Mug*"

No color, material, style, lighting, or background constraints.

STRUCTURED (Improved)

"*Professional studio photography of a white ceramic Coffee Mug, minimalist aesthetic, centered composition, soft cinematic lighting, white solid background, 8k resolution, photorealistic, sharp focus.*"

1.2 Evaluation Setup

- **CLIP Score**: `openai/clip-vit-base-patch32` — measures semantic alignment between text prompt and generated image. Computed on seed-42 images.
- **SSIM**: Pairwise Structural Similarity Index averaged across all 3 seeds (42, 123, 999) per product. Higher = more consistent visual structure.

2. Full Quantitative Results

ID	Product	Base CLIP	Imp CLIP	ΔCLIP	Base SSIM	Imp SSIM	ΔSSIM
1	Women Summer Dress	0.249	0.252	+0.003	0.262	0.344	+0.082
2	Running Shoes	0.283	0.227	-0.056	0.100	0.177	+0.077
3	Coffee Mug	0.315	0.301	-0.014	0.460	**0.732**	+0.272
4	Leather Backpack	0.305	0.301	-0.004	0.614	0.258	-0.356
5	Wrist Watch	0.268	0.279	+0.011	0.088	0.191	+0.103
6	Hoodie	0.302	0.301	-0.001	0.414	0.421	+0.007
7	Sneakers	0.294	0.260	-0.034	0.170	0.349	+0.179
8	Handbag	0.289	0.279	-0.010	0.422	0.540	+0.118
9	Smart Speaker	0.272	0.284	+0.012	0.498	**0.716**	+0.218
10	Wireless Earbuds	0.307	0.306	-0.001	0.570	**0.691**	+0.121
	Average	**0.288**	**0.279**	-0.009	**0.360**	**0.442**	+0.082

3. Analysis & Key Findings

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CLIP Alignment Preserved

Average CLIP decreased by only -0.009 (negligible). SDXL identifies the core object correctly regardless of prompt length — adding 25+ control tokens does not confuse subject recognition.

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Consistency Improved +22.8%

SSIM improved from 0.360 to 0.442 on average. Environmental anchoring (white background, cinematic lighting, centered composition) constrains the diffusion sampler's degrees of freedom.

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Best Gains: Simple Geometry

Coffee Mug (+59%), Smart Speaker (+44%), and Sneakers (+105%) benefited most. Simple, smooth-surfaced objects respond best to structured prompts.

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Material Tokens Effective

Keywords like "ceramic," "leather," and "mesh" visibly changed surface textures in outputs. Attribute-level control at the material domain is highly effective in SDXL.

4. Failure Cases & Limitations

📦 Leather Backpack — SSIM Regression

SSIM dropped from **0.614 → 0.258**. The structured prompt activated different sub-concepts per seed (buckles vs. zippers vs. straps), creating structural divergence despite individually good images.

💡 **Lesson: Multi-component products need explicit positional tokens (e.g., "single front pocket, no zippers")**

👟 Running Shoes — Low SSIM Overall

Both baseline (**0.100**) and improved (**0.177**) remain low. Footwear with laces, mesh, and sole geometry has high natural variance across seeds.

💡 **Lesson: ControlNet or img2img conditioning required for geometric precision**

⌚ Wrist Watch — Fine Detail Loss

Dial numbers and watch hands are inconsistent. SDXL's 1024×1024 resolution is insufficient for sub-millimetre rendering of fine mechanical details.

💡 **Lesson: Super-resolution post-processing (ESRGAN) needed for accessory detail**

📊 Running Shoes — Low Baseline

The simple title "Running Shoes" is ambiguous — the model generates everything from high-tops to low-profile, various color schemes, and different viewing angles per seed.

💡 **Lesson: Even structured prompts cannot fully compensate for inherent model variance on complex objects**

⚖️ Key Trade-off: Control vs. Diversity

Higher SSIM (consistency) naturally reduces creative variety. For **e-commerce catalogs** this is a desirable trade-off — brand consistency matters more than diversity. For **artistic generation**, baseline prompts produce more interesting variation.

5. Conclusions

- ✅ Structured prompt engineering **measurably improves cross-seed consistency (+22.8% SSIM)** for e-commerce images.
- ✅ CLIP alignment is **preserved** — adding control tokens does not degrade object identification.
- ✅ Pipeline is **fully automated and reproducible** — 60 images generated from CSV metadata via one-click Colab execution.
- ⚠️ Products with complex geometry (footwear, multi-pocket bags) require additional control mechanisms beyond text prompting.
- 🚀 Future work: ControlNet for pose/angle control; real Amazon metadata for 100+ product scalability test; ESRGAN super-resolution for accessory detail.